

MATRIX SYSTEM ARCHITECTURE

Targeted Module Delivery & Quadrant Refinement Blueprint

Core System Logic: MATRIX is an automated, student-exclusive platform designed to evaluate 1st-year BSIT students using Richard Elmore's 4 quadrants. The core loop strictly tests the student, calculates quadrant scores, identifies the **weakest** learning area, and continuously feeds targeted Python modules to improve that exact weakness until a specific passing threshold is met. (Note: Teacher portals are excluded from scope).

1. Student Portal (1st-Year BSIT Interface)

A. The Diagnostic Assessment Page

- **The Questionnaire:** A 20-to-30 item survey to plot the student on Richard Elmore's 4 quadrants (Hierarchical, Distributed, Individual, Collective).
- **Helper Text (Panel Requirement):** Every question must have a contextual helper text below it to clarify the logic.

Example: "Do you prefer the professor to give you a manual to read, or do you want to experiment with the code yourself until it works?"

- **Scoring Engine:** Backend logic that calculates and securely stores the coordinates.

B. Personal Dashboard & Thresholds

- **Weakness Highlighter:** A prominent UI element explicitly stating the target.
Example: "Your current weakest quadrant is [Individual Learning]. Your targeted focus is to improve independent coding skills."
- **Threshold Progress Indicator:** A visual gauge set to a specific number (e.g., 80%). As students pass modules, the gauge fills up. Hitting the target flags the learning modality as "Effective."

C. Targeted Module Delivery (Python Focus)

- **Weakness-First Routing:** The algorithm actively suggests and unlocks modules specifically designed to train the student's lowest-scoring quadrant.
- **Foundational Content:** Modules focus on 1st-year essentials: (1) Variables & Data Types, (2) Control Logic (If/Else), (3) Loops, (4) Functions.

D. Varied Assessment Types (Multiple Instances)

- **Multiple Learning Instances:** Students can take tests multiple times to track progression. Failed tests trigger a new, distinct instance of the assessment.
- **Type 1 - Concept Check:** Multiple Choice for basic theory.
- **Type 2 - Code Tracing:** Fill-in-the-blanks where students input missing syntax.
- **Type 3 - Logic Sequencing:** Drag-and-drop to unscramble lines of Python code.
- **Type 4 - Output Prediction:** Text input where the student predicts the console output.

E. Analytics & "The 6 Graphs" Report

- **CRITICAL REQUIREMENT:** Every graph generated must have the exact percentage numbers visibly printed directly on the charts.
- **1. Initial Quadrant Plot (Radar Chart):** Starting scores in all 4 Elmore quadrants.
- **2. Weakness Identification (Bar Graph):** Lowest quadrant score vs. class average.
- **3. Module Attempt Instances (Line Graph):** Number of tries to pass each module.
- **4. Assessment Type Success (Pie Chart):** Breakdown of best-performing assessment types (e.g., 40% Drag-and-Drop, 60% Multiple Choice).
- **5. Weakness Progression (Line Graph):** Trajectory of their weakest quadrant score over time.
- **6. Threshold Hit Rate (Gauge Chart):** Final score crossing the "Effective" target threshold.

2. Admin & Researcher Portal

A. Module & Assessment Management

- **Dynamic Upload Interface:** A backend page to upload Python lessons (PDFs/videos) and attach the four different types of varied assessments.
- **Questionnaire Editor:** An interface to edit the Richard Elmore survey questions and input the mandatory "Example" helper text beneath them.

B. Logic & Threshold Controls

- **Adjustable Threshold Slider:** A backend setting to manually adjust the "passing" percentage target, proving to the panel that the system is fully configurable.
- **Quadrant Weighting Matrix:** Backend math storage easily accessible for Behavioral Professionals or Psychometricians to validate.

C. ISO 25010 Data Hub

- **System-Wide Progress:** Aggregate percentage graphs showing the improvement of all 134 respondents.
- **Raw Data Export:** A one-click export feature that dumps all student scores, assessment instances, and graph percentages into a clean Excel sheet for rapid manuscript integration.